QUALIFICATION STANDARD
AND PLAN (QSP)

Marine System Engineer (MS Eng) (00345)
Head of Department (HOD) (AISE)

MITE ID
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OJT – 114809

Regular Force: MS Eng (00345)
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RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.00		04 Jul 05	DMTE
1.01		07 Jul 15	NPTG
1.01		20 Oct 15	NPTG
2.0	Adopt new RCN QS and TP format	16 June 23	CO HMCS VENTURE

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Personnel and Training Group (CNPTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the MSEO Job Based Occupational Specification dated 4 November 2004 A-PD-055-002/PP-001;

3. This QSP publication is effective on receipt and supersedes previous QSP revision 1.01 dated 20 October 2015. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened in 2007 at CFNES, CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened in 2007 at CFNES, CFB Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Marine System Engineering (MS Eng) to perform their duties as a Marine System Engineering Officer (MSEO). Possession of this qualification will allow personnel to become a Head Of Department (HOD).

GRANTING OF QUALIFICATION

2. A trainee will achieve the qualification AISE, Marine Systems Engineering Head of Department when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. N/A

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

   a. Be MS Eng Basic Occupation (AIRY) qualified

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001 Manage Departmental Personnel

1. Performance: Manage Departmental Personnel

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References

    a. Denied:

        (1) Supervision; and

    b. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

    a. Advise on Operational Limitations due to Personnel Shortages to include evaluating the impact of personnel limitations on operational requirements by determining departmental requirements to meet the operational schedule and addressing personnel shortfalls.

    b. Managing Individual Training and Education by:

        (1) Reviewing career progression and requirements for all occupations in the department;

        (2) Reviewing and maintaining training documentation;

        (3) Conducting individual interviews/counselling;

        (4) Liaising with career management and training personnel to achieve required training; and

(5) Monitoring and conducting training.

c. Provide divisional support for military personnel by:

(1) Maintaining divisional notes;

(2) Producing administrative staff work;

(3) Counselling of subordinates' personal issues that impact on job performance;

(4) Acting as Assisting/Attending Officer;

(5) Evaluating subordinates through the CFPAS/PACE process; and

(6) Acknowledging superior performers via honours and awards programs and accelerated promotions.

d. Manage operational training including command team weapons system certification and by ensuring refresher training and team training are current to meet operational requirements by:

(1) Scheduling exercises to meet CRRs;

(2) Developing DC policy and procedures; and

(3) Liaising with Training Chief/PO to determine training requirements.

4. Reference: As per Annex A.

5. Task Numbers: BT5, BT10, BT15, CT10, CT15, CT50, CT55, ET15.

6. Remarks: Nil.

PO 002 Liaise with Scientific establishments and Industry

1. Performance: Liaise with Scientific establishments and Industry

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Direction from Formation authorities.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), the officer shall effectively liaise with laboratories, industries and establishments to support naval engineering such as:

   a. Defence Research and Development Canada (DRDC);

   b. NETE;

   c. QETE; and

   d. OEM/FSR.

4. Reference: As per Annex A.

5. Task Numbers: AT30.

6. Remarks: Nil.

PO 003 Manage Departmental Financial Resources

1. Performance: Manage Departmental Financial Resources.

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine;

        (2) References;

        (3) Departmental Budget;

        (4) FAA Section 32, Section 34 (Expenditure Management); and

        (5) Qualification.

    b. Denied:

        (1) Supervision.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), the officer shall:

    a. Manage the departmental budget and ensure compliance with procurement and contracting policy to include:

    b. Completion of the Expenditure Management course;

    c. Reviewing policy;

    d. Reviewing budget allocation;

    e. Establishing priorities;

    f. Tracking expenditures; and

    g. Ensuring appropriate items are properly accounted for and documented.

4. Reference: As per Annex A.

5. Task Numbers: MT40, MT50.

6. Remarks: Nil.

PO 004 Manage Shipboard Risk

1. Performance: Manage Shipboard Risk

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Unit Environmental Officer Qualification.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), Relevant safety and environmental directives, the officer shall:

   a. Promote environmental awareness onboard ship by being a member of the Ship's Environmental Committee;

   b. Manage Interdepartmental safety procedures including ensuring compliance with general safety policies and procedures by being a member of the Ship's general safety committee;

   c. Perform the duties of the UEnvO:

      (1) Planning, reviewing and implementing pollution control measures; and

      (2) Implementing and monitoring energy conservation measures.

   d. Perform the duties of the head of the Fuel Spill Response Team;

   e. Ensure compliance with policies/ governing procurement, stowage, handling, disposal and training of hazardous materials.

4. Reference: As per Annex A.

5. Task Numbers: AT40, FT30, FT40, FT35, FT5, FT15, AT45.

6. Remarks: Nil.

PO 005 Prepare Technical Documentation

1. Performance: Prepare Technical Documentation

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Supervision.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), the officer shall:

    a. Complete Routine Engineering Reports such as:

        (1) PELs;

        (2) Combat Readiness Requirement (CRR) Report; and

        (3) Monthly Maintenance Reports.

    b. Review and maintain technical directives, orders and records such as:

        (1) EOTIs;

        (2) EOTMs; and

        (3) Captain's Ship Book.

    c. Prepare maritime technical policy by reviewing, drafting and commenting upon documents such as:

        (1) MARLANTORDS/MARPACORDS;

        (2) VSTIs;

(3) Fleet SOPs; and

(4) Technical Bulletins.

d. Review Fire Reports.

4. Reference: As per Annex A.

5. Task Numbers: AT5, DT105, GT15, FT 20.

6. Remarks: Nil

PO 006 Participate in Interdepartmental Operational Planning

1. Performance: Participate in Interdepartmental Operational Planning

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), the officer shall:

   a. Participate in planning ship's operations, program and/or routine to determine optimum system configurations by liaising with other heads of department to determine departmental requirements for the program;

   b. Liaise with CSEO to assess impact of marine systems on combat systems; and

   c. Ensure ship's mobility state is achieved to carry out projected operations.

4. Reference: As per Annex A.

5. Task Numbers: GT70, DT30, DT125, DT(copied from 135 CS) MS.

6. Remarks: Nil.

PO 007 Participate in Signature Management Program

1. Performance: Participate in Signature Management Program

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine;

        (2) References; and

        (3) Signature Management Organization.

    b. Denied:

        (1) Supervision.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), the officer shall:

    a. Perform duties of Ship's Noise Control Officer; and

    b. Participate in the ship's signature management organization.

4. Reference: CFCD 109, NAVORD 3395-1.

5. Task Numbers: IT40, DT55, DT125.

6. Remarks: Nil.

PO 008 Manage Shipboard Configuration

1. Performance: Manage Shipboard Configuration

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Approved As-Fitted drawings.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall control configuration management onboard ship by managing configuration changes to include:

   a. Implementing approved configuration changes;

   b. Initiating Engineering Change Proposals; and

   c. Preventing unauthorized changes to configured items.

4. Reference: As per Annex A.

5. Task Numbers: HT35, LT40, LT55.

6. Remarks: Nil.

PO 009 Manage Test and Trials Program

1. Performance: Manage Test and Trials Program

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Trial Agendas; and

      (4) Operational Schedule.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall manage the ship's tests and trials programme to include:

   a. Developing a test and trials programme through liaison with applicable technical and operational authorities;

   b. Reviewing and commenting on trial agendas;

   c. Conducting tests and trials;

   d. Analyzing trial reports and advising command of results; and

   e. Taking necessary action to rectify defects.

4. Reference: As per Annex A.

5. Task Numbers: KT35, KT20, KT10, KT5.

6. Remarks: Nil.

PO 010 Manage Readiness

1. Performance: Manage Readiness

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

   a. Maximize marine systems availability within the stated notice for power to include:

      (1) Assessing technical readiness and advising command on mission readiness by:

         (a) Monitoring the status of marine systems;

         (b) Assessing pre-sail checks;

         (c) Conducting engineering rounds;

         (d) Reviewing periodic equipment diagnostics and performance checks;

         (e) Assessing Departmental personnel readiness;

         (f) Preparing an OPDEF Summary message;

         (g) Analyzing the effects of equipment degradation and assessing impact on other systems;

(h) Verifying the currency of maintenance and operating documentation; and

(i) Verifying spare parts inventory.

(2) Directing maintenance activities.

b. Maximize marine systems availability at sea by:

(1) Advising command on mission readiness by:

(a) Assessing technical readiness;

(b) Monitoring of marine systems;

(c) Conducting engineering rounds;

(d) Reviewing logs and state boards;

(e) Assessing Departmental personnel readiness;

(f) Preparing an OPDEF summary report; and

(g) Analyzing the effects of equipment degradation and assessing impact on other systems.

(2) Directing maintenance activities.

c. Assess status of DC states; and

d. Advise on Shipboard Degrees of Readiness.

4. Reference: As per Annex A.

5. Task Numbers: DT5, DT10, DT15, DT20, DT25, DT30, DT35, DT50, DT 65 NCS, DT85.

6. Remarks: Nil.

PO 011 Manage Defect Rectification

1. Performance: Manage Defect Rectification

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) Operational Schedule.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall direct and supervise repair action due to equipment failures to include:

   a. Receiving defect reports from operators and maintainers;

   b. Investigating cause, nature and extent of equipment defects and failures;

   c. Assessing impact of equipment failures on operations and advising command on operational limitations under various system operating modes;

   d. Determining requirements for equipment and repair team safety and isolating system faults;

   e. Identifying requirements and methods of obtaining spares;

   f. Planning emergency repairs by taking command priorities and directing repair teams as appropriate; and

   g. Drafting and managing OPDEFs.

4. Reference: As per Annex A.

5. Task Numbers: DT40 (copied from NCS Eng Occ Spec), DT95, IT10, DT120, GT75, DT20, DT45, GT35.

6. Remarks: Nil

PO 012 Manage Battle Damage

1. Performance: Manage Battle Damage

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall direct DCHQ1, DCHQ2 and section base teams in action and emergency situations to include:

   a. Receiving damage reports;

   b. Comparing command capability priorities with reported damage;

   c. Take command priorities and direct repair parties appropriately; and

   d. Advise command on impact of damage, equipment/systems casualties and spares.

4. Reference: As per Annex A.

5. Task Numbers: ET25, ET30, ET10.

6. Remarks: Nil.

PO 013 Respond to non-routine operations

1. Performance: Respond to non-routine operations

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

   a. Direct a Fuel Spill Response team to include:

      (1) Ensuring duty watch, individual and team training are current; and

      (2) Ensuring necessary equipment is stocked and/or maintained.

   b. Direct shipboard response to engineering emergencies.

4. Reference: As per Annex A.

5. Task Numbers: ET(TBD) MS, ET5.

6. Remarks: Nil.

PO 014 Manage Docking

1. Performance: Manage Docking

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

   a. Perform docking calculations and interpret the docking plan prior to liaising with the docking officer/dockmaster;

   b. Determine the ship's condition for docking/undocking to include:

      (1) Determining the requirements to change trim to meet the docking condition; and

      (2) Preparing the docking/undocking temporary memorandum.

   c. Prepare ship for docking/undocking to include:

      (1) Meeting departmental requirements of docking/undocking memorandum; and

      (2) Ensuring all other departments comply with the docking/undocking memorandum requirements.

   d. Prepare docking/undocking reports through liaison with the docking officer/dockmaster/FTA; and

   e. Conduct docking/undocking.

4. Reference: As per Annex A.

5. Task Numbers: IT15, IT30, DT90, IT25, IT20, IT35.

6. Remarks: Nil.

PO 015 Manage Ship's Tiered Readiness Program (TRP)

1. Performance: Manage Ship's Tiered Readiness Program (TRP)

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate repair facility support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall project manage the departmental responsibilities during a TRP to include:

   a. Managing equipment readiness; and

   b. Managing personnel training.

4. Reference: As per Annex A.

5. Task Numbers: TBD.

6. Remarks: Nil.

PO 016 Manage Formation Level Maintenance Activities

1. Performance: Manage Formation Level Maintenance Activities

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems;

      (4) Appropriate repair facility support; and

      (5) Appropriate Formation support.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

   a. Prepare and staff first line O&M documentation to include:

      (1) Identifying deficiencies in maintenance schedules through the UCR process;

      (2) Developing new maintenance procedures and techniques;

      (3) Evaluating new maintenance procedures and techniques; and

      (4) Reviewing and approving DRMIS Maintenance Notifications and associated stores demands.

   b. Use naval maintenance management information systems to include:

      (1) Identifying deficiencies in maintenance schedules through the UCR process;

(2) Developing new maintenance procedures and techniques;

(3) Evaluating new maintenance procedures and techniques; and

(4) Reviewing and approving DRMIS Maintenance Notifications and associated stores demands.

c. Liaise with Formation staff with respect to priority to include:

(1) Identifying/referring maintenance tasks beyond ship's staff capability; and

(2) Prioritizing maintenance action to include:

(a) Reviewing corrective maintenance;

(b) Reviewing current preventive maintenance; and

(c) Reviewing outstanding preventive maintenance.

d. Monitor contract work to include:

(1) Contract progress against scheduled milestones;

(2) Technical compliance with contracts; and

(3) Reviewing contract deliverables for compliance.

e. Participate in the Test Equipment Maintenance Management Information System (TEMMIS) program.

4. Reference: As per Annex A.

5. Task Numbers: HT40, LT80, HT30, GT TBD, LT75, GT40, DT110, HT20, GT45, DT115, HT25.

6. Remarks: Nil.

PO 017 Manage a Short Work Period (SWP)

1. Performance: Manage a Short Work Period (SWP)

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate repair facility support; and

      (4) Appropriate Formation support.

   d. Denied:

      (1) Supervision.

   e. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall project manage a SWP to include:

   a. Preparing for the short work period;

   b. Managing the short work period; and

   c. Completing the short work period.

4. Reference: As per Annex A.

5. Task Numbers: TBD.

6. Remarks: Nil.

PO 018 Manage ADM(Mat) Level Maintenance Activities

1. Performance: Manage ADM(Mat) Level Maintenance Activities

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

   a. Plan a refit/docking work period to include pre-refit/DWP trials and surveys;

   b. Prepare and staff 3rd line O&M documentation;

   c. Monitor contract work to include:

      (1) Contract progress against scheduled milestones;

      (2) Observation reports for defects/anomalies during refits/DWPs/Ship construction;

      (3) Technical compliance with contracts; and

      (4) Reviewing contract deliverables for compliance.

   d. Providing support to Quality Assurance Representatives to include:

(1) Monitoring contract progress against scheduled milestones;

(2) Drafting observation reports for defects/anomalies during refits/DWPs/Ship construction;

(3) Ensuring technical compliance with contracts; and

(4) Reviewing contract deliverables for compliance.

4. Reference: As per Annex A.

5. Task Numbers: JT50, JT5, MT20, MT30, JT30, NT120, MT25, NT150, MT35, NT155.

6. Remarks: Nil.

PO 019 Manage Ship's Stability

1. Performance: Manage Ship's Stability

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Trim and Stability computer program; and

      (4) Manual of Trim and Stability.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

   a. Assess ship's stability to include observing stability indicators such as draught marks, inclinometer readings, etc,

   b. Manage the disposition of fluids/cargo to maintain ship's trim and stability; and

   c. Maintain ship's trim, stability and stresses within safe limits.

4. Reference: As per Annex A.

5. Task Numbers: PT5, PT20, PT15.

6. Remarks: Nil.

PO 020 Manage Liquid Load

1. Performance: Manage Liquid Load

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Supervision.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs (para 4), relevant safety and environmental directives, the officer shall:

   a. Ensure safe handling and stowage of fluids to include providing advice on the safe handling, stowage and transfer of petroleum products;

   b. Control liquid transfer to include:

      (1) Preparing a fuelling/defuelling/liquid transfer plan;

      (2) Directing the embarkation, transfer and storage of fluids;

      (3) Advising command on the quality of fuel; and

      (4) Preparing a final report on the ship's condition on completion of the fuelling evolution.

   c. Maintain a fuel log.

4. Reference: As per Annex A.

5. Task Numbers: S5, S10, S15.

6. Remarks: Nil.

PO 021 Manage Departmental Systems

1. Performance: Manage Departmental Systems

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Supervision.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: The MSEO shall manage departmental systems in accordance with references by:

    a. Managing Main Propulsion Systems to include:

        (1) Monitoring start-up, running and shutdown of systems;

        (2) Reviewing system logs and operating records;

        (3) Monitoring diagnostic tests;

        (4) Monitoring equipment performance checks; and

        (5) Advising command on system status.

    b. Managing Hull Systems to include:

        (1) Monitoring start-up, running and shutdown of systems;

        (2) Reviewing system logs and operating records;

        (3) Monitoring diagnostic tests;

        (4) Monitoring equipment performance checks;

(5) Advising command on system status;

(6) Directing survey of damaged/wasted hull structures; and

(7) Examining hull structures/systems.

c. Managing Auxiliary Systems to include:

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

d. Managing Control Systems to include:

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

e. Manage Electrical Systems to include:

(1) Monitoring start-up, running and shutdown of systems;

(2) Reviewing system logs and operating records;

(3) Monitoring diagnostic tests;

(4) Monitoring equipment performance checks; and

(5) Advising command on system status.

f. Manage departmental operational requirements to include:

(1) Providing orders;

(2) Monitoring equipment;

(3) Monitoring personnel;

(4) Managing documentation; and

(5) Advising command.

4. Reference: As per Annex A.

5. Task Numbers: OT135, OT140, OT145, OT150, OT155, JT35, JT40, GT50, GT55, GT60, DT6.

6. Remarks: Nil.

TRAINING PLAN

CHAPTER 3– TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. On the job training

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 365 training days.

REQUIRED COURSE STAFFING

3. There are no required staffing as this is an on the job training course conducted on an HMC Ship

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

   a. Nil

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

   a. Optimum – is 1 as this training is to be conducted while onboard a ship as an Assistant Marine Systems Engineering Officer

   b. Maximum Load – is 1 as this training is to be conducted while onboard a ship as an Assistant Marine Systems Engineering Officer

   c. Minimum – is 1 as this training is to be conducted while onboard a ship as an Assistant Marine Systems Engineering Officer

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-20. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – Manage Departmental Personnel

EO 001.01 Advise on Operational Limitations due to Personnel Shortages

1. Performance: Advise on Operational Limitations due to Personnel Shortages

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall evaluate the impact of personnel limitations on operational requirements to include:

    a. Determining departmental requirements to meet the operational schedule by:

        (1) Reviewing AER to include MARTECH qualifications;

        (2) Review ship's program; and

        (3) Liaise with Chief Eng regarding Watch and Station Bill.

    b. Addressing personnel shortfalls by:

        (1) Brief XO on personnel readiness; and

        (2) Draft a personnel shortage message.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 001 Manage Departmental Personnel

EO 001.02 Manage Individual Training and Education

1. Performance: Manage Individual Training and Education

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall manage individual training and education by:

   a. Reviewing career progression and requirements for all occupations in the department;

   b. Reviewing and maintaining training documentation;

   c. Conducting individual interviews/counselling;

   d. Liaising with career management and training personnel to achieve required training; and

   e. Monitoring and conducting training to include:

      (1) Prioritizing training requirements;

      (2) Organizing and assessing training opportunities; and

      (3) Participating in pre-boards and qualification boards.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 001 Manage Departmental Personnel

EO 001.03 Provide Divisional support for military personnel

1. Performance: Provide Divisional support for military personnel

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall perform as a Divisional Officer to include:

   a. Maintaining divisional notes;

   b. Producing administrative staff work;

   c. Counselling of subordinates' personal issues that impact on job performance. Subjects to include:

      (1) Administrative measures;

      (2) Drugs, alcohol and harassment;

      (3) Financial issues;

      (4) Sexual harassment and/or misconduct;

      (5) Advise on alternate career opportunities including commissioning plans, occupational transfers and release; and

      (6) Personal relationships.

   d. Acting as Assisting/Attending Officer;

e. Evaluating subordinates through the PAR process; and

f. Acknowledging superior performers via honours and awards programs and accelerated promotions.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 001 Manage Departmental Personnel

EO 001.04 Manage operational training

1. Performance: Manage operational training

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall manage operational training to include:

   a. Participating in in damage control organization team training;

   b. Ensuring refresher training and team training are current to meet operational requirements by:

      (1) Scheduling exercises to meet CRRs;

      (2) Developing DC policy and procedure through liaison with Sea Training staff and training establishments; and

      (3) Liaising with Training Chief/PO to determine training requirements.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 002 Liaise with Scientific establishments and Industry

EO 002.01 Liaise with Scientific establishments and Industry

1. Performance: Liaise with Scientific establishments and Industry

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, the officer shall demonstrate comprehensive knowledge of the roles and capabilities of laboratories, industries and establishments, and the procedures to request their assistance, to support naval engineering. These units can include:

   a. Defence Research and Development Canada (DRDC);

   b. NETE;

   c. QETE; and

   d. OEM/FSR.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 003 Manage Departmental Financial Resources

EO 003.01 Manage departmental budget / ensure compliance with policy

1. Performance: Manage departmental budget / ensure compliance with policy

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, the officer shall manage the departmental budget and ensure compliance with procurement and contracting policy to include:

   a. Completion of the Expenditure Management course;

   b. Reviewing policy to include unit level direction on budget management;

   c. Reviewing budget allocation;

   d. Establishing priorities;

   e. Tracking expenditures;

   f. Ensuring appropriate items are properly accounted for and documented; and

   g. Letting a contract in a foreign port.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A

7. Remarks: Nil.

PO 004 Manage Shipboard Risk

EO 004.01 Promote environmental awareness onboard ship

1. Performance: Promote environmental awareness onboard ship

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall promote environmental awareness onboard ship by being a member of the Ship's Environmental Committee.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 004 Manage Shipboard Risk

EO 004.02 Manage interdepartmental safety procedures

1. Performance: Manage interdepartmental safety procedures

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall manage interdepartmental safety procedures to include:

    a. Ensuring compliance with general safety policies and procedures by:

        (1) Assisting in the completion of a SPDET;

        (2) Completing monthly safety rounds; and

        (3) Completing CF98s and DND663s.

    b. Participating in the Ship’s general safety committee.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 004 Manage Shipboard Risk

EO 004.03 Perform as Unit Environmental Officer

1. Performance: Perform as Unit Environmental Officer

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall perform the duties of the UEnvO to include:

   a. Completion of the Unit EnvO course;

   b. Conducting shipboard environmental training;

   c. Attending formation level environmental meetings and briefings;

   d. Performing as the Ship's Assistant UEnvO;

   e. Reviewing environmental incident messages;

   f. Implementing pollution control measures;

   g. Implementing and monitoring energy conservation measures;

   h. Initiating investigations into environmental incidents to include:

      (1) Drafting environmental incident messages; and

      (2) Requirements for further investigation.

   i. Reviewing results of investigation and take appropriate actions.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 004 Manage Shipboard Risk

EO 004.04 Perform the duties of the head of the Fuel Spill Response Team

1. Performance: Perform the duties of the head of the Fuel Spill Response Team

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall perform the duties of the head of the Fuel Spill Response Team to include:

   a. Ensuring duty watch, individual and team training are current;

   b. Ensuring necessary equipment is stocked and/or maintained; and

   c. Carrying out fuel spill response exercises.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 004 Manage Shipboard Risk

EO 004.05 Compliance with policies/procedures governing HazMat

1. Performance: Ensure compliance with policies/procedures governing handling, stowage and disposal of hazardous materials

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall ensure compliance with policies/procedures governing procurement, stowage, handling, disposal and product-specific training for hazardous materials.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 005 Prepare Technical Documentation

EO 005.01 Complete Routine Engineering Reports

1. Performance: Complete Engineering Reports

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall complete routine engineering reports such as:

    a. PELs;

    b. Combat Readiness Requirement (CRR) Report;

    c. Monthly Maintenance Report.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 005 Prepare Technical Documentation

EO 005.02 Review and maintain technical directives, orders and records

1. Performance: Review and maintain technical directives, orders and records

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall review and maintain technical directives, orders and records such as:

   a. EOTIs:

   b. EOTMs;

   c. Captain's Ship Book;

   d. Lifting Appliances database;

   e. Halocarbon log;

   f. Boiler tube history;

   g. Pressure vessel database;

   h. QA Live file; and

   i. Supercession.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 005 Prepare Technical Documentation

EO 005.03 Prepare maritime technical policy

1. Performance: Prepare maritime technical policy

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall prepare maritime technical policy by reviewing, drafting and commenting upon documents such as:

   a. NAVORDS/MARLANTORDS/MARPACORDS;

   b. VSTIs;

   c. Fleet SOPs;

   d. Technical Bulletins;

   e. CFTOs;

   f. NEM; and

   g. NaMMS.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 006 Participate in Interdepartmental Operational Planning

EO 006.01 Participate in planning ship's operations, program and/or routine

1. Performance: Participate in planning ship's operations, program and/or routine

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall participate in planning ship's operations, program and/or routine to include:

   a. Liaising with other heads of department to provide input for the long and short term programs (i.e. SAR, SUBSAR, long term deployment, force protection, TRP, Missile EX, ASWEX) to include:

      (1) Test and Trials requirements;

      (2) Training requirements; and

      (3) Maintenance requirements.

   b. Liaising with other heads of department to provide engineering support by reacting to the ship's long and short term programs (i.e. SAR, SUBSAR, long term deployment, force protection, TRP, Missile EX, ASWEX) with respect to:

      (1) Mission requirements;

      (2) Maintenance requirements;

      (3) Training requirements; and

(4) Optimizing system configuration.

c. Participating in:

(1) HODs and CHODs;

(2) XO's Morning Prayers;

(3) In Port Training (IPT); and

(4) CO's Evening Brief.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 006 Participate in Interdepartmental Operational Planning

EO 006.02 Assess impact of marine systems on combat systems

1. Performance: Liaise with CSEO to assess impact of marine systems on combat systems

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall liaise with the CSEO to assess the impact on combat systems of the loss or limitation of the following marine systems:

   a. Power Generation and Distribution;

   b. HVAC;

   c. LP/HP Air;

   d. Chilled Water;

   e. Firemain/Sea Water Services;

   f. Fitted Firefighting equipment; and

   g. Hydraulics.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 007 Manage Signature Management Program

EO 007.01 Participate in Ship's Signature Management

1. Performance: Participate in Ship's Signature Management

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall participate in the Ship's Signature Management Organization to include applying signature management policies and procedures by:

    a. Conducting signature management rounds and ranging;

    b. Reviewing results from rounds and ranging;

    c. Briefing the CO on signature management rounds and ranging; and

    d. Reporting and rectifying defects through OPDEFs and corrective maintenance.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 008 Manage Shipboard Configuration

EO 008.01 Control configuration management onboard ship

1. Performance: Control configuration management onboard ship

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Approved As-Fitted drawings

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall control configuration management onboard ship to include managing configuration changes by:

   a. Implementing approved configuration changes by liaising with formation authorities;

   b. Initiating Engineering Change Proposals including supporting documentation such as UCRs and SOCDs; and

   c. Preventing unauthorized changes to configured items.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 009 Manage Test and Trials Programme

EO 009.01 Manage Ship's tests and trials programme

1. Performance: Manage Ship's tests and trials programme

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Trial Agendas Operational Schedule

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall manage Ship's tests and trials programme to include:

   a. Developing a test and trials programme through liaison with applicable technical and operational authorities;

   b. Reviewing and commenting on trial agendas;

   c. Conducting tests and trials such as:

      (1) Full power trials;

      (2) Steering gear trials;

      (3) Deep dive trials; and

      (4) Load Bank trials.

   d. Analyzing trial reports and advising command of results; and

   e. Taking necessary action to rectify defects.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 010 Manage Readiness

EO 010.01 Maximize marine systems availability within the stated NFP

1. Performance: Maximize marine systems availability within the stated notice for power

2. Conditions:

a. Given:

(1) MSE Dept in a ship or submarine; and

(2) References.

b. Denied:

(1) Nil.

c. Environment:

(1) Alongside;

(2) At sea; and

(3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall maximize marine systems availability within the stated notice for power to include:

a. Assessing technical readiness and advising command on mission readiness by:

(1) Monitoring the status of marine systems;

(2) Assessing pre-sail checks;

(3) Conducting engineering rounds to include:

(a) Ensuring departmental spaces are secured for sea;

(b) Ensuring departmental spaces are free of hazards; and

(c) Ensuring departmental spaces contain proper fire fighting/damage control/safety equipment.

(4) Reviewing periodic equipment diagnostics and performance checks;

(5) Assessing Departmental personnel readiness;

(6) Preparing an OPDEF summary report;

(7) Verifying the currency of maintenance and operating documentation; and

(8) Verifying departmental stores inventory.

b. Directing maintenance activities.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 010 Manage Readiness

EO 010.02 Maximize marine systems availability at sea

1. Performance: Maximize marine systems availability at sea

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall maximize marine systems availability at sea to include:

   a. Assessing technical readiness and advising command on mission readiness by:

      (1) Monitoring the status of marine systems;

      (2) Conducting engineering rounds:

         (a) Ensuring departmental spaces are secured for sea;

         (b) Ensuring departmental spaces are free of hazards; and

         (c) Ensuring departmental spaces contain proper fire fighting/damage control/safety equipment.

      (3) Managing fire prevention inspections;

      (4) Reviewing periodic equipment diagnostics and performance checks;

      (5) Reviewing logs and state boards;

(6) Assessing Departmental personnel readiness;

(7) Preparing an OPDEF summary report; and

(8) Analyzing the effects of equipment degradation and assessing impact on other systems.

b. Directing maintenance activities.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 010 Manage Readiness

EO 010.03 Assess status of DC states

1. Performance: Assess status of DC states

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, (page 6.) relevant safety and environmental directives, the officer shall assess status of DC States to include:

   a. Demonstrating a complete understanding of the personnel/technical requirements of all watertight and gastight integrity states to include:

      (1) X-RAY;

      (2) YANKEE;

      (3) ZULU;

      (4) YANKEE-BRAVO; and

      (5) ZULU-ALPHA.

   b. Advising command on the impact (i.e. personnel/equipment) of maintaining watertight/gastight integrity;

   c. Advising command on the impact of deviating from standard watertight/gastight integrity conditions (i.e. ongoing maintenance); and

   d. Advising command on watertight/gastight integrity conditions to correspond with ship's operations (i.e. littoral operations).

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 010 Manage Readiness

EO 010.04 Advise on Shipboard Degrees of Readiness

1. Performance: Advise on Shipboard Degrees of Readiness

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, the officer shall:

   a. Manage the employment of departmental personnel to meet all Shipboard Degrees of Readiness using the following resources:

      (1) Watch and Station Bill;

      (2) Chief Eng and Cox'n to resolve conflicts in the WSB; and

      (3) Requirements of special parties.

   b. Advise command on the capabilities and limitations on departmental personnel to meet the required degrees of readiness.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 011 Manage Defect Rectification

EO 011.01 Direct and supervise repair action due to equipment failures

1. Performance: Direct and supervise repair action due to equipment failures

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall direct and supervise repair action due to equipment failures to include:

   a. Receiving defect reports from operators and maintainers;

   b. Investigating cause, nature and extent of equipment defects and failures;

   c. Assessing impact of equipment failures on operations and advising command on operational limitations under various system operating modes to include:

      (1) Environmental factors (weather, sea state, etc);

      (2) Ship’s services;

      (3) Operational conditions;

      (4) Modes of operations;

      (5) Operator action at time of fault;

      (6) Equipment diagnostics;

      (7) Operational and maintenance history; and

(8) State of related equipment.

d. Determining requirements for equipment and repair team safety and isolating system faults;

e. Identifying requirements and methods of obtaining spares to include:

(1) EMREQs;

(2) TRANREQs;

(3) High Priority Stores Requirements;

(4) PIFs;

(5) LCMM liaison;

(6) Forward Logistics Site; and

(7) Manufacture from original drawings.

f. Planning repairs by taking command priorities and directing repair teams to include:

(1) Inputs and concerns such as:

(a) Discussing with operators and maintainers;

(b) Utilizing appropriate manuals;

(c) Determining operational considerations to rectify fault, including consultation with command and OOW/ORO/Dived Watch Leader; and

(d) Determining activities/schedules of other departments that could have an impact.

(2) Availability of spares and other resources; and

(3) Results of liaison with other departments.

g. Directing departmental personnel to rectify faults; and

h. Performing as the departmental OPDEF coordinator to include drafting and managing OPDEFs.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 012 Manage Battle Damage

EO 012.01 Describe the Ship's Damage Control Organization

1. Performance: Describe the Ship's Damage Control Organization

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) DC Manuals.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, (page 6.) relevant safety and environmental directives, the officer shall describe the DC organization at Action, Emergency and Emergency Flying Stations to include:

   a. Composition;

   b. Tools/handbooks/repair material;

   c. Location;

   d. Ship wide resources;

   e. Control;

   f. Duties and responsibilities (initial responses/follow up actions);

   g. Incident boards/state boards;

   h. Means of communication; and

   i. Differences in reactions to Action Stations/Emergency Stations/Emergency Flying Stations.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 012 Manage Battle Damage

EO 012.02 Direct DCHQ1/DCHQ2/Section Base Teams

1. Performance: Direct DCHQ1/DCHQ2/Section Base Teams

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) DC Manuals.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall direct DCHQ1/DCHQ2/Section Base Teams to include:

   a. Determining status of SBTs when closed up to include:

      (1) Location of personnel;

      (2) Verification of DC equipment;

      (3) Readiness of personnel;

      (4) Communication checks;

      (5) Reporting status of machinery; and

      (6) Reporting status to command.

   b. Assessing marine systems readiness to include:

      (1) Updating equipment status;

      (2) Verifying nature of threat/situation with command; and

(3) Reporting readiness to command.

c. Performing the duties of DCO to include:

(1) Advising command of the damage control situation;

(2) Coordinating the work of SBTs and other elements of the DC organization;

(3) Controlling deliberate changes to the ship's stability; and

(4) Advising command regarding priority for electrical power.

d. Directing SBTs to include:

(1) Relevant threat/situation information;

(2) Priorities of response to damage/failures;

(3) Direct action to be conducted such as:

(a) Casualty/emergency power;

(b) Location of emergency/damage;

(c) Smoke Control and Ventilation;

(d) Isolating power to various equipment; and

(e) Deliberate use of fitted DC systems.

(4) Change location of SBTs/DCHQs as required.

e. Monitoring equipment state/damage to include:

(1) Receiving reports from SBTs and Ch Eng;

(2) Receiving reports from other sources (Casualty Clearing/ERT);

(3) Monitoring relevant DC alarms and equipment alerts; and

(4) Consulting with command to prioritize activities for action or emergency stations.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 012 Manage Battle Damage

EO 012.03 Manage Damage Control training

1. Performance: Manage Damage Control training

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) DC Manuals.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall direct DCHQ1/DCHQ2/Section Base Teams to include:

   a. Operational team training such as:

      (1) DCOTT;

      (2) Table-top/Comms exercises;

      (3) Helo Firefighting;

      (4) SWOAD;

      (5) SBT training; and

      (6) Force Protection.

   b. Alongside training such as:

      (1) Duty Watch training;

      (2) Ship Initiated Emergency Exercises; and

(3) IPT.

c. At sea training such as:

(1) Ship initiated Emergency Exercises;

(2) Table-top/Comm exercises;

(3) DC Skill at hand; and

(4) Battle problems.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 013 Respond to non-routine operations

EO 013.01 Direct the Fuel Spill Response Team

1. Performance: Direct the Fuel Spill Response Team

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) Class EMS manual; and

      (3) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall direct the Fuel Spill Response Team to include:

   a. Taking appropriate actions to stop the release and contain the spill;

   b. Evacuating the area;

   c. Cleaning up the spill site;

   d. Advising command on clean-up progress and impact;

   e. Reporting, in accordance with formation policy, the disposal action taken, probable consequences of the spill and the success of the clean-up operation; and

   f. Conducting an investigation.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 013 Respond to non-routine operations

EO 013.02 Respond to Shipboard Engineering Emergencies

1. Performance: EO Performance Statement (verb, object & qualifier)

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Class specific Engineering Emergency Drills

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall respond to Shipboard Engineering Emergencies to include:

   a. Planning the conduct of engineering drills to include:

      (1) Determining CRRs/ITE requirements;

      (2) Liaising with the OPSO for scheduling; and

      (3) In conjunction with MARTECH Training PO, generate a 1 hour drill period to include a minimum of 4 drills.

   b. Pre- Engineering Drill Action:

      (1) Conduct Pre-drill meeting with assessment team, including safety numbers;

      (2) Review drills and brief expected actions/evolution of drills;

      (3) Brief members on expectations/actions to facilitate drills; and

(4) Discuss actions to take in the event of a real incident or emergency occurs.

c. Conducting and assessing engineering drills to include:

(1) Determining effectiveness of the EOOW and the Engineering Watch;

(2) Determining accuracy and completeness of reports passed to the OOW;

(3) Determining the proper conduct of the drill; and

(4) Assess the proper conduct of the drill.

d. Performing as the MSEO during drills to include:

(1) Accepting reports;

(2) Advising command; and

(3) Planning follow-up actions and repairs.

e. Post Engineering Drill Action:

(1) Confirm Engineering Plant is reconfigured to normal state; and

(2) Debrief Engineering Watch.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 014 Manage Docking

EO 014.01 Describe the types of docks and the caps and lims of each

1. Performance: Describe the types of docks and the capabilities and limitations of each

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of the types of docks and the capabilities and limitations of each to include:

   a. Docking terminology to include:

      (1) Bilge blocks;

      (2) Bilge cribs;

      (3) Keel block towers;

      (4) Breast shores;

      (5) Bilge shores;

      (6) Cribbed blocks;

      (7) Docking condition; and

      (8) Sueing.

   b. Graving docks to include:

(1) Describing the typical configuration including pumping requirements and gate considerations;

(2) Describing the typical docking consideration, including:

(a) Dock floor capacity;

(b) Effects of tides; and

(c) Use of sonar/rudder pits.

c. Syncrolifts to include:

(1) Describing the typical configuration including basic principles of operation and certification requirements; and

(2) Describe typical docking considerations, including maximum load and load per unit length.

d. Floating dry docks to include:

(1) Describing the typical configuration including basic principles of operation, certification requirements and the ability to dock with trim on blocks; and

(2) Describing typical docking considerations, including maximum load.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 014 Manage Docking

EO 014.02 Describe the occasions and regulations for docking

1. Performance: Describe the occasions and regulations for docking

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of:

   a. The occasions for docking to include:

      (1) Refit/DWP/EDWP;

      (2) Emergency docking; and

      (3) Non-scheduled docking.

   b. The regulations for docking to include:

      (1) Responsibilities of the MSEO;

      (2) Responsibilities of the Docking Officer;

      (3) Responsibilities of the Dockmaster; and

      (4) Differences in responsibilities of Docking Officer or Dockmaster.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 014 Manage Docking

EO 014.03 Describe the preparation required for docking

1. Performance: EO Performance Statement (verb, object & qualifier)

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of the preparations for docking to include:

   a. Reviewing and confirming the docking plan for suitability, compatibility and validity to include:

      (1) Blocks (locations and locating);

      (2) Shore (locations);

      (3) Shore services;

      (4) Tide requirement;

      (5) Dock plan rotation (IAW Captain's Ship Book);

      (6) Trim and draught requirements;

      (7) Propeller position requirements;

      (8) Sonar dome positioning requirements; and

      (9) Hull fitting position requirements.

b. Inspecting the dock with the Docking Officer and Dock Master for debris, hazards, and conformity of docking blocks to plan;

c. Liaising with appropriate authorities to include QAWC, FMF, FTA and the contractor to include:

(1) Determining and meeting ship/submarine condition for docking to include:

(a) Engineering considerations;

(b) Importance of being free from ammunition; and

(c) Defuelling requirements.

(2) Identifying available dock services; and

(3) Reviewing contract specification to identify work that might impact on block loading and ship's/submarine's structure.

d. Coordinating requirements with other Heads of Department;

e. Advising the CO on technical aspects of the docking operation;

f. Establishing a burning and welding control plan;

g. Establishing a weight control plan;

h. Establishing a docking organization to include:

(1) Manning of all spaces below the waterline;

(2) Working of valves;

(3) Communications;

(4) Reports; and

(5) Changing machinery status.

i. Explaining stability considerations to include:

(1) Ship's/submarine's hydrostatic properties from measured drafts and predict the ship's/submarine's LCG;

(2) Procedures for calculating critical drafts; and

(3) Stability of a vessel docking with trim.

j. Preparing a COTM; and

k. Preparing and delivering a docking brief.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 014 Manage Docking

EO 014.04 Describe weight controls and safety considerations while docked and the regulations/preparations for undocking

1. Performance: Describe weight controls and safety considerations while docked and the regulations/preparations for undocking

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall:

   a. Demonstrate a detailed knowledge of weight control and safety considerations on a docked ship/submarine:

   b. Prepare for undocking to include:

      (1) Undocking certificate to include:

         (a) Explaining the contents of the CF 523; and

         (b) Preparing a sample CF 523.

      (2) Determining the effect of weight changes on trim;

      (3) Liaising with appropriate authorities (QAWC, FMF, FTA and contractor) to ensure ship's/submarine's condition prior to undocking;

      (4) Coordinating requirements with other department heads;

(5) Establishing an undocking organization to include:

(a) Manning of all spaces below the waterline;

(b) Working of valves;

(c) Communications; and

(d) Reports.

(6) Preparing COTM.

c. Inspect the dock prior to undocking with the docking officer and dockmaster;

d. Advise the CO on technical aspects of the undocking operation; and

e. Give detailed verbal reports during and on completion of the undocking evolution.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 014 Manage Docking

EO 014.05 Conduct a docking/undocking

1. Performance: Conduct a docking/undocking

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of:

   a. Support services (i.e. shore services, tugs, divers);

   b. Duration of docking evolution;

   c. Environmental considerations (tide, wind);

   d. Safety considerations; and

   e. Division of responsibility.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 015 Manage Ship's Tiered Readiness Program (TRP)

EO 015.01 Project manage departmental responsibilities during a TRP

1. Performance: Project manage departmental responsibilities during a TRP

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate Formation and repair facility support; and

      (4) TRP Alert letter/OPORD

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall project manage departmental responsibilities during a TRP by:

   a. Managing equipment readiness to meet TRP objectives to include:

      (1) Readiness inspections;

      (2) Post DWP/refit trials;

      (3) Pre-SRI alongside trials;

      (4) SRIs;

      (5) Fuelling;

      (6) Readiness trials at sea;

      (7) Defect rectification/EC installation SWP;

      (8) Re-trials; and

(9) Work-ups.

b. Managing personnel to meet TRP objectives to include:

(1) Ensuring that refresher and team training are up to date; and

(2) Planning leave periods.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 016 Manage Formation Level Maintenance Activities

EO 016.01 Prepare and staff first line O&M documentation

1. Performance: Prepare and staff first line O&M documentation

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall prepare and staff first line O&M documentation using the naval maintenance information system to include:

   a. Identifying deficiencies in maintenance schedules through the UCR process;

   b. Developing new maintenance procedures and techniques, and drafting documentation for review by higher authority;

   c. Evaluating new maintenance procedures and techniques; and

   d. Reviewing and approving DRMIS Maintenance Notifications and associated stores demands.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 016 Manage Formation Level Maintenance Activities

EO 016.02 Liaise with Formation staff with respect to priority

1. Performance: Liaise with Formation staff with respect to priority

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall liaise with formation staff with respect to priority to include:

   a. Identifying/referring maintenance tasks beyond ship's staff capability; and

   b. Prioritizing corrective maintenance, current preventive maintenance and outstanding preventive maintenance, taking into consideration:

      (1) Outstanding OPDEFs;

      (2) Ship's program;

      (3) Safety requirements;

      (4) SUBSAFE requirements;

      (5) Equipment availability; and

      (6) Resource availability.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 016 Manage Formation Level Maintenance Activities

EO 016.03 Monitor contract work

1. Performance: Monitor contract work

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall monitor contract work to include:

   a. Contract progress against scheduled milestones;

   b. Technical compliance with contracts; and

   c. Reviewing contract deliverables for compliance.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 016 Manage Formation Level Maintenance Activities

EO 016.04 Participate in the TEMMIS programme

1. Performance: Participate in the Test Equipment Maintenance Management Information System (TEMMIS) programme

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Current naval maintenance management information systems; and

      (4) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall identify TEMMIS requirements.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 017 Manage a Short Work Period (SWP)

EO 017.01 Project Manage a SWP

1. Performance: Project Manage a SWP

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Appropriate Formation and repair facility support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall project manage a SWP to include:

   a. Preparing for the short work period by:

      (1) Receiving SWP Alert message;

      (2) Ensuring all Work Requests (CM, PM and ECs) for SWP are entered into the naval maintenance information system;

      (3) Drafting SWP Response Message to include prioritizing work taking into consideration:

         (a) Outstanding OPDEFs;

         (b) Ship’s program;

         (c) Safety requirements;

         (d) SUBSAFE requirements;

         (e) Equipment availability; and

(f) Resource availability.

(4) Receiving work request priority list;

(5) Sending completed work request priority list to FMF and N37;

(6) Attending the Ship's Contract Acceptance Meeting; and

(7) Briefing command on SWP issues.

b. Managing the short work period to include:

(1) Attending progress review meetings;

(2) Addressing internal and external issues and conflicts such as:

(a) Personnel availability;

(b) Ship's activities;

(c) Equipment availability;

(d) Ship's services;

(e) Cranes;

(f) Divers;

(g) Other ships;

(h) QHM;

(i) RF work priorities; and

(j) Safety.

(3) Briefing command on the progress of SWP.

c. Completing the short work period to include:

(1) Attending SWP wrap-up meeting;

(2) Providing FMF feedback of overall effectiveness;

(3) Prioritizing outstanding work; and

(4) Briefing command on results of SWP.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 018 Manage ADM(Mat) Level Maintenance Activities

EO 018.01 Plan a refit/docking work period

1. Performance: Plan a refit/docking work period

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall plan a refit/docking work period to include:

   a. IAW refs relevant safety and environmental directives, the officer shall plan a refit/docking work period to include:

   b. Reviewing the Refit Alert letter;

   c. Reviewing the SSRWC;

   d. Identifying particular defects and provide a detailed work description in the DND 650s;

   e. Scheduling and completing required pre-refit surveys, trials and inspections with the assistance of formation engineering, FMF and class desk contractors in accordance with the OPSKED;

   f. Submitting DND 650s to the MRSL development agent on an "as completed" basis, not awaiting the completion of all DND 650s;

g. Providing feedback to class desk on DND 650 priority list and accepted PMRS list;

h. Managing de-fuelling of ship;

i. Managing departmental de-storing;

j. Managing departmental preparation for transit to and from contractors yard such as:

(1) Receipt of transit pack; and

(2) Personnel requirements for force protection.

k. Participating in the Supply Overhaul Assistance Program (SOAP); and

l. Developing a personnel employment and training plan.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 018 Manage ADM(Mat) Level Maintenance Activities

EO 018.02 Monitor contract work

1. Performance: Minor contract work

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, the officer shall monitor contract progress against scheduled milestones to include:

   a. Ensuring technical compliance with contracts;

   b. Reviewing contract deliverables for compliance;

   c. Reviewing Observation reports for defects/anomalies during refits/DWPs/Ship construction;

   d. Managing set-to-work requirements to include:

      (1) Reviewing set-to-work schedule;

      (2) Providing feedback on set-to-work schedule to class desk through QAWC; and

      (3) Providing technical expertise/operators as required.

   e. Ensuring system operability for undocking and acceptance to include:

(1) Completion and set-to-work of docking dependent work; and

(2) Completion and set-to-work for final acceptance.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 018 Manage ADM(Mat) Level Maintenance Activities

EO 018.03 Provide support to Quality Assurance Representatives

1. Performance: Provide support to Quality Assurance Representatives

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, the officer shall provide support to Quality Assurance Representatives to include:

   a. Augmenting the on-site QAR team by supplying the services of certain technical and support personnel in accordance with detailed direction to be promulgated by the class desk and formation staff;

   b. Ensuring qualified personnel are available to operate equipment during set-to-work period;

   c. Liaising with QAR to minimize risk associated with critical milestones to include:

      (1) Government supplied materiel;

      (2) Contractor furnished materiel;

      (3) Special tools and test equipment (SPTATE); and

      (4) Docking dependent work.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 019 Manage Ship's Stability

EO 019.01 Manage Ship's Stability

1. Performance: Manage Ship's Stability

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References;

      (3) Appropriate ADM(Mat) support; and

      (4) Appropriate Formation support.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall:

   a. Assess ship's stability to include observing stability indicators such as draught marks, inclinometer readings, etc;

   b. Manage the disposition of fluids/cargo to maintain ship's trim and stability; and

   c. Maintain ship's trim, stability and stresses within safe limits to include:

      (1) Deep departure;

      (2) Operational light;

      (3) Docking;

      (4) Icing; and

      (5) Calculating and setting diving trim.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 020 Manage Liquid Load

EO 020.01 Ensure safe handling and stowage of fluids

1. Performance: Ensure safe handling and stowage of fluids

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall provide advice on and ensure safe handling, stowage and transfer of petroleum products such as:

    a. Gasoline;

    b. GP-357-M;

    c. GP-36-M;

    d. Synthetic Lube Oil; and

    e. MIL 9000.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 020 Manage Liquid Load

EO 020.02 Control liquid transfer

1. Performance: Control liquid transfer

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine;

      (2) References; and

      (3) Trim and Stability computer program.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall control liquid transfer to include:

   a. Preparing a fuelling/defuelling/liquid transfer plan to include Liquid Husbandry;

   b. Directing the embarkation, transfer and storage of fluids to include:

      (1) F-76;

      (2) F-44 including aircraft operations (HIFR);

      (3) Potable water;

      (4) De-ionized/Feed water;

      (5) Salt water ballast; and

      (6) Bilge water.

   c. Advising command on the quality of fuel;

d. Preparing a final report on the ship's condition on completion of the fuelling evolution;

e. Maintain a fuel log; and

f. Perform the duties of the Fuelling Officer.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 021 Manage Departmental Systems

EO 021.01 Describe Main Propulsion and Ancillary Systems

1. Performance: Describe Main Propulsion and Ancillary Systems

2. Conditions:

    a. Given:

        (1) MSE Dept in a ship or submarine; and

        (2) References.

    b. Denied:

        (1) Nil.

    c. Environment:

        (1) Alongside;

        (2) At sea; and

        (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of main propulsion and ancillary systems to include:

    a. Purpose;

    b. Main components including basic drawing;

    c. Operation (basic);

    d. Configuration (modes of operation);

    e. Capabilities, Limitations and Redundancies;

    f. Safety;

    g. Personnel and maintenance requirements;

    h. Abnormal operations; and

    i. Interfacing/Integration with other systems.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 021 Manage Departmental Systems

EO 021.02 Describe Auxiliary Systems

1. Performance: Describe Auxiliary System

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of auxiliary systems to include:

   a. Purpose;

   b. Main components including basic drawing;

   c. Operation (basic);

   d. Configuration (modes of operation);

   e. Capabilities, Limitations and Redundancies;

   f. Safety;

   g. Personnel and maintenance requirements;

   h. Abnormal operations; and

   i. Interfacing/Integration with other systems.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 021 Manage Departmental Systems

EO 021.03 Describe Control systems

1. Performance: Describe Control Systems

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of control systems to include:

   a. Purpose;

   b. Main components including basic drawing;

   c. Operation (basic);

   d. Configuration (modes of operation);

   e. Capabilities, Limitations and Redundancies;

   f. Safety;

   g. Personnel and maintenance requirements;

   h. Abnormal operations; and

   i. Interfacing/Integration with other systems.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 021 Manage Departmental Systems

EO 021.04 Describe Electrical systems

1. Performance: Describe Electrical Systems

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall demonstrate a detailed knowledge of electrical systems to include:

   a. Purpose;

   b. Main components including basic drawing;

   c. Operation (basic);

   d. Configuration (modes of operation);

   e. Capabilities, Limitations and Redundancies;

   f. Safety;

   g. Personnel and maintenance requirements;

   h. Abnormal operations; and

   i. Interfacing/Integration with other systems.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 021 Manage Departmental Systems

EO 021.05 Describe Hull systems

1. Performance: Describe Hull Systems

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs, relevant safety and environmental directives, the officer shall:

   a. demonstrate a detailed knowledge of hull systems to include:

      (1) Purpose;

      (2) Main components including basic drawing;

      (3) Operation (basic);

      (4) Configuration (modes of operation);

      (5) Capabilities, Limitations and Redundancies;

      (6) Safety;

      (7) Personnel and maintenance requirements;

      (8) Abnormal operations; and

      (9) Interfacing/Integration with other systems.

   b. examine hull structures; and

c. direct survey of damaged/wasted hull structures.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

PO 021 Manage Departmental Systems

EO 021.06 Manage MSE Departmental Requirements

1. Performance: Describe Hull Systems

2. Conditions:

   a. Given:

      (1) MSE Dept in a ship or submarine; and

      (2) References.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Alongside;

      (2) At sea; and

      (3) Under all conditions day and night.

3. Standard: IAW refs,the trainee shall manage departmental requirements by:

   a. Managing administrative requirements to include:

      (1) Providing written and oral orders to direct the safe and efficient operation of marine systems;

      (2) Monitoring the capability of departmental personnel for safe and efficient operation of systems;

      (3) Managing technical documentation;

      (4) Advising command by describing machinery:

         (a) availability;

         (b) operation;

         (c) capability;

         (d) limitations;

         (e) configuration;

(f) repair requirements;

(g) personnel; and

(h) safety.

b. Managing operational requirements of Main Propulsion, Auxiliary, Hull, Electrical and Control Systems to include:

(1) Monitoring the flash-up, operation and shutdown of equipment and systems by:

(a) observing procedures;

(b) conferring with personnel;

(c) reviewing system logs and records;

(d) conducting pre-sail checks;

(e) conducting scheduled equipment checks;

(f) conducting rounds; and

(g) initiating defect rectification by:

i. identifying system deficiencies;

ii. analyzing effects of equipment degradation; and

iii. assessing impact on other systems and ship operations.

c. Managing equipment casualty scenarios of Main Propulsion, Auxiliary, Hull, Electrical and Control equipment by:

(1) Briefing command on a system's technical problem or failure to include:

(a) situation;

(b) possible causes;

(c) engineering and operational impact;

(d) options to include:

i. work arounds;

ii. repair;

iii. risk assessment;

iv. risk mitigation;

v. supply considerations; and

vi. timeline.

(e) recommendations;

(f) implementation plan; and

(g) requirement to liaise with subject matter experts and external agencies.

d. Administrative and documentation requirements.

4. Teaching Points: Nil.

5. Total EO Time: Throughout 12 months of training time.

6. Training and Learning Aids: N/A.

7. Remarks: Nil.

QUALIFICATION STANDARD ANNEXS

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP. As there was no references in the original QSP before the rewrite, the table is here to be completed.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01		
A02		
A03		
A04		
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01		
B02		
B03		
C		**Civilian Publications and Manuals that relate to this Qualification**
C01		
C02		
D		**Other Publications and Manuals that relate to this Qualification**
D01		
D02		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments

ANNEX C – QS SCALAR

N/A

ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	Marine Systems Engineering Assistant Head of Department Qualification Standard and Plan			
Location of Board	CFNES OT Div			
Dates	8 Feb to 25 Feb 05			
Board Composition				
	Surname	Initials	Rank	Unit
Chair	Stuart	R.A	LCdr	CFNES
SME	Harrison	S.C.W	LCdr	CFNES
SME	Bradford	T.L.	LCdr	CANFLTPAC
SME	Gregg	S.C	LCdr	MOG5HQ
DMTE Rep	Shirriff	T.	LCdr	DMTE
SME	Budden	H.L.	Lt(N)	HMCS IROQUOIS
SME TDO	Higdon	T.S.W	Lt(N)	NDQAR
Secretary	Checketts	A.	Lt(N)	DMTE
NTDSS Operator	Waller	R.D	Lt(N)	FMF CB
	Jones	R.	LS	CFFS(E)

INTRODUCTION

1. The Chairman opened the proceedings by welcoming everyone to the QSPWB and provided a brief outline of the objective, scope, constraints and process for the QSPWB. The QSPWB members briefly introduced themselves and Lt(N) Checketts presented a detailed overview of CFITES and the QSP process.

2. It was assumed that the candidates would arrive for the A/HOD training having completed their Basic Occupation Qualification (BOQ) and would therefore be at the requisite skill level to perform the tasks expected of the A/MSEO or A/CSEO job in the respective Integrated Occupation Specification.

3. The QSPWB was tasked to develop the HOD training for both NCS and MS Eng officers. The initial focus of the QSPWB was to examine the common tasks between both occupations and to group them in order to maximize efficiency. This involved a detailed review of the Integrated Occupation Specifications (references A and B). This review resulted in a list of recommended changes to both Integrated Occupation Specifications. These recommended changes can be categorized in three areas:

a. Wording changes/Incompatibilities between MS Eng and NCS Eng (Annex A);

b. Skill level changes related to specific tasks (Annex B); and

c. Knowledge level changes (Annex C)

4. Due to the nature of the recommended changes to the Integrated Occupation Specifications and on the advice of the DMTE representative, the QSPWB assumed that the recommended changes to the Integrated Occupation Specifications would be accepted. This meant that the development of the HOD training used the tasks, skills and knowledges from references A and B in addition to the changes recommended at the Annexes as the baseline.

TRAIN/NO TRAIN DECISIONS

5. As noted above, it was assumed that the candidates arriving to carry out the A/HOD training would have the requisite skill level associated with the A/MSEO or A/CSEO job in the respective Integrated Occupation Specification. This was important as the QSPWB focussed only on tasks where there was a skill level increase for the task from the assistant job (A/MSEO or A/CSEO) to the HOD job (MSEO or CSEO). The following items were determined to be NO TRAIN tasks:

a. FT30 MS Perform as Unit/Command General Safety Officer - As noted in Annex A, this item is only in the MS Eng IOS. It was recommended that this task be removed from the MS Eng IOS as it is not specifically related to the job of an MS Eng, rather it is a task that could be assigned to any qualified person within a unit. Further, there is specific UGSO training given elsewhere;

b. Duty Area O – Marine/Combat Systems – As noted in Annex A, it is recommended that each task in the MS Eng IOS be split into two tasks, “Manage” and “Assess” to mirror the NCS Eng IOS. The “Assess” tasks are no train since there is no increase in skill level from the A/MSEO or A/CSEO job to the MSEO or CSEO job, recognizing the fact that the MS or NCS Eng will have achieved a level 3 for the “Assess” tasks at completion of the BOQ (i.e. phase VI);

ISSUES DISCUSSED

6. The QSPWB discussed many issues in the development of the QSP. Many of the issues related to the Integrated Occupation Specifications and are captured in Annexes A to C. There were other issues that did not fall directly into these categories. These issues are captured below and, where necessary, present the reasons that a particular decision/recommendation was made on the issue.

a. Conducting A/HOD training on a different class of vessel than BOQ –

There were two issues related to this point –what level of systems knowledge is expected of the A/HOD and should additional time be given to a trainee that does change platforms after BOQ:

i. Level of systems knowledge – It was agreed by the QSPWB that the Manage Marine/Combat Systems Performance Objective (PO 020 for NCS Eng and PO 021 for MS Eng) addressed the expected system level knowledge for the A/HOD, and

ii. Additional time – It was agreed that no additional time would be automatically given to a trainee that changes class of vessel after BOQ. This would be addressed in Chapter 1, specifically stating that the CO has the authority to grant a 25% time extension (in accordance with MARCORD 9-28) and, a change of class of vessel after BOQ is a valid reason to grant this extension;

b. The QSPWB agreed that the Naval Technical Manager Refresher (formerly TORC) is a useful course for the A/HOD to attend prior to joining their ship as the A/HOD;

c. Monitoring of training progression – The discussion centered on the issue of who should be monitoring progression. The overall responsibility for this lies with DMTE. Some ideas that were presented included having CANFLT involved (i.e. GTO / F4 Eng) or having CFNES / CFFSE involved. The QSPWB recommended that DMTE retain the overall responsibility for monitoring trainee progression with the assistance of formation agencies. The specifics of who will perform this on each coast would have to be worked out appropriately. The QSPWB recommended that, given the fact that the ships report to the Commander of the CANFLT organizations (CCFL/CCFP), the GTO / F4 Eng would be the appropriate coastal OPI;

d. NBCO for NCS Eng submariners – It was determined that this topic was outside the scope of the QSPWB;

e. Divisional Officer Duties – It was noted that in the BOQ OJPRs, there is a PO to “Lead Personnel” that is common to the MS and NCS Eng, however, there is not a specific EO in the afloat phase to act as a divisional officer. The OJPRs state in the instructions to the MSEO/CSEO that the candidate (i.e. phase VI) shall be given divisional officer duties, but this is not reflected in the OJPR. It was recommended that an afloat EO be added to the QSPs for the BOQs;

f. Manage Readiness States – A lengthy discussion on this topic was

held. The decision was made to break this into three elements:

i. Manage Readiness Alongside,
ii. Manage Readiness At Sea, and
iii. Tiered Readiness Program;

g. Manage Marine/Combat Systems – Managing the marine/combat systems was determined to be an all encompassing requirement – all the other performance objectives provided supporting tasks, skills and knowledges to performing as the head of the department. Adding a “Perform the duties of the CSEO/MSEO” Enabling Objective to each of the Manage Systems Performance Objectives emphasized this;

h. It was recommended that, prior to completing the HOD qualification, the A/CSEO complete the Unit RadHaz Officer course and the A/MSEO complete the Unit Environmental Officer course;

i. Risk Analysis - Knowledge of how to conduct risk analysis needs to be increased further through formal training. The QSPWB recommended that formalized lectures on the current risk analysis processes and policies such as SUBSAFE be added to the BOQ ashore coursing. In addition, the knowledges associated with this item (K0745 NCS / K0795 MS) should be increased to 2 for the A/MSEO / A/CSEO. At the MSEO / CSEO level these knowledges should increase to 3 and continue at 3 for post HOD positions;

j. Reference CD – There was a discussion on the merits of providing a reference CD for the A/HOD. It was decided that providing a CD (or other media) with the references for the AHOD was not beneficial for the training. There were two reasons for this decision. The first was that the QSPWB agreed that the ability to find, interpret and understand the references is an important skill to learn and secondly, the references change on a regular basis;

k. Number of Trainees embarked – The AHOD position is an established billet with a valid job – that of being the Assistant HOD. It should be noted that this position comes with the responsibility to carry out OJT while performing the duties required of the Assistant HOD. The trainee completing their BOQ (phase VI) is there primarily to carry out their training. If there is no AHOD, the phase VI trainee should not be used primarily to fill the gap, as it will impact on their training. It was recommended that no more than one A/CSEO and one A/MSEO be posted to each ship/submarine. It was acknowledged that there may be situations where an overlap occurs, however, the amount of time for the overlap should be minimized;

l. MS Eng BOQ, EO 403.2 – It was noted that this enabling objective has no mention of the PROTECTEUR Class propulsion control system. This needs to be examined and if phase VI candidates continue to be sent to PROTECTEUR Class ships, suitable amendments to the MS Eng OJPR should be made;

m. Guidance to the ship (CO/XO/HOD) for A/HOD training –This point was discussed and consideration was given to:

   i. Having the CCA send a letter to the ship with specific guidance for the HOD and a "contract" to be signed by the HOD, acknowledging their role; or
   ii. Having a coastal representative (i.e. CFNES/CFFSE) visit the ship when an AHOD is posted in to clarify the training requirements and address any questions of concerns that the ship may have; or
   iii. Inclusion of the specific guidance for the HOD training in the "DMTE Guidance to COs" annual letter.

   This guidance will be included in the front of the OJT package and it is recommended that it be included in the DMTE drafted "Commanding Officers Training Responsibilities" annual letter;

n. NCS Eng BOQ, EO 411.4, para 10.b. - Recommend this be re-written to indicate that the trainee will participate in HAZMAT cleanup in various positions (ie. Member of Cleanup team, dressing area assistant, OSC, command plotter);

o. In the NCS Eng BOQ OJPR, EO 414.5, para 10c ("Plan and conduct a DMP/DWP/SWP") should be deleted as this is an AHOD requirement. Note that this may be a transcription error, as this statement is not in the equivalent MS Eng BOQ EO (EO 13.5); and

p. It is recommended that the theory of heavy lift ships should be added to the MS Eng BOQ (EO 10.1).

7. There was a lengthy discussion on assessment. Historically, the HOD qualification has been assessed through an oral board after completion of the OJT package. The QSPWB went through the process of determining how to train these qualifications and it was determined that the best method was through OJT. Within the chapter 4 that was developed by the QSPWB, which forms the core material for the OJT package, each of the enabling objectives has a "Test Details" section that dictates what the candidate must do to complete a particular enabling objective.

8. It was further determined that completion of an oral board after the OJT package has been completed is the best method of formally assessing a candidate. Having a board with the content controlled by the CCA and the board conducted by CFNES for MARLANT candidates and by CFFSE for MARPAC candidates will ensure that a common standard is applied to all trainees.

OUTSTANDING ACTION REQUIRED

9. The core content of the QSP was completed by the QSPWB. The CCA now has all of this material and is progressing the remaining administrative work to finalize the QSP.

RECOMMENDED WAY AHEAD

10. The completed QSP will be forwarded to DMTE for approval of chapters 1 and 2. Once approved, CFNES as CCA will approve the remainder of the QSP. It is anticipated that the revised training package for both MS and NCS Eng AHODs can be ready to be distributed to the trainees commencing in APS 05.

CONCLUSION

11. I consider this QSPWB to have been a success. The team was able to maintain focus for the three week period. The QSPs created will not result in a radical change in A/HOD training, however, the NTO community will have a current and, in the opinion of the QSPWB members, a valid, relevant and more applicable guide to training our future heads of department.

// original signed by //
R.A. Stuart
Lieutenant Commander
Chairman, Combined MS and NCS Eng HOD QSPWB

Annex A Integrated Occupation Specification - Wording Changes/Incompatibilities
Annex B Integrated Occupation Specification – Skill Value Changes
Annex C Integrated Occupation Specification - Knowledge Value Changes

Annex A - Integrated Occupation Specifications - Wording Changes/Incompatibilities

- BT005 – Reword from “manning shortfalls” to “Personnel Shortages”
- BT NEW: “Counsel Performance” should be added to both IOS
- CT10 – MS wording should mimic NCS – i.e. add “assess”
- CT15 MS – wording should mimic CT 15 NCS – i.e. “Plan/conduct/assess …”
- DT15 – Reword to include “Operational” Degrees of Readiness
- DT NEW – Create a new requirement titled “Advise on Ship’s Tiered Readiness Program”
- DT35: Reword from “Advise on technical readiness” to “Advise on mission readiness”
- DT40 MS should be added to NCS and DT40 NCS should be added to MS
- DT65/70 MS needs to be added to NCS
- MS Staff O needs to know about NCS systems but the reciprocal for NCS is not necessary (CT35, DT 35 MS)
- Insert DT5 MS into NCS
- DT65 “Assess pre-sail checks” not included in MS. It needs to be added.
- Amend DT115 NCS to read “Identify requirements for and methods of obtaining spare parts”. For the MS IOS, insert this statement to replace DT120 MS.
- There is no “Participate in planning ship’s ops/programme/routine” (DT125 NCS) for MS. This should be added to the MS IOS.
- DT130 NCS – This task will be divided into two separate tasks: “Maximize combat systems availability within the stated notice for power” and “Maximize combat systems availability at sea”. These two tasks also need to be inserted in the MS IOS
- DT135 NCS – a similar task should be created for MS. The QSPWB decided on “ Assess the impact of MSE systems on Combat Systems”
- ET 5 NCS - “Respond to missile emergencies” is not applicable to all classes of ship. The task should be reworded to “Respond to ammunition emergencies”
- ET 30 MS should be reworded to match ET 20 NCS
- ET35 MS can be deleted because it is encompassed in DT45
- There should be a “Direct fuel spill response team” in the MS IOS to mimic ET30 NCS “Direct HAZMAT Cleanup team”. This task was added as ET35 MS.
- FT5 MS – Reword from “Prepare” to “Ensure compliance”
- FT5 – The task should be reworded to match the EMS SOP to read “governing procurement, stowage, handling, disposal and training of hazardous materials”
- FT30 MS - Performing as the UGSO is listed as an MS task. UGSO is not an occupation specific job, therefore it should not be included in the IOS. It should be replaced with FT40 NCS “Ensure compliance with general safety policy and procedures”.
- GT50 – Change the wording from “Conduct” to “Monitor”
- GT95 NCS should be inserted in the MS IOS, although the task is not applicable for this QSP

- GT NEW – “Participate in the Test Equipment Maintenance Management Information System (TEMMIS) program” should be added to the MS IOS
- IT NEW - “Project Manage a Short Work Period” should be added to both IOS
- Adjust the wording of JT5 MS to match JT5 NCS. Remove “Short Work Period” from the description because a new task will be created to focus solely on SWPs
- JT25 MS does not appear in the NCS IOS and should be added. However, this task is not applicable to this QSP
- Duty Area K should be changed in NCS to mimic the MS IOS.
- LT 75 MS is absent from the NCS IOS and needs to be added
- LT 85 NCS is not present in the MS IOS and should be added. However, this task is not applicable to this QSP
- MT40 NCS needs to be added to the MS IOS
- MT45 NCS and MT50 NCS are absent from the MS IOS and should be added.
  The QSPWB also recommends that these two tasks belong in Duty Area A
- Duty Area N differs between the NCS and MS IOS. The tasks listed in the MS IOS are more applicable and should be copied to the NCS IOS with the exception of NT100, which should not be a task for the Asst/EO
- In section O in the MS IOS, each task includes “Manage/Analyze”. The format is different in the NCS IOS where the two words are separated into different tasks. Since Analyzing systems does not result in an increase in knowledge, the section O tasks for MS should mirror the format of the tasks in the NCS IOS
- OT55 and OT60 NCS should be reworded from “Under/Above Water Warfare systems” to “Under/Above Water Sensor systems”
- OT80 NCS – The mirror task “Manage/assess Combat Systems” should be included in the MS IOS. This task is trained at the NEI level, so it is not applicable to this QSP
- OT140 MS – The task should be reworded to remove “Main” from “Main Auxiliary Systems”
- RT25 NCS – This task should be removed because it is not appropriate to have the Asst/CSEO involved in Ship/Design/Construction
- K1065 MS/ K1125 NCS – The knowledge should be reworded from “there” to “their”

Annex B - Integrated Occupation Specifications - Skill Value Changes

- AT40 A/MSEO should be a 2
- AT45 A/MSEO should be a 2
- DT40 A/MSEO, Jnr Eng and Jnr Eng SO should be a 2 to match the NCS IOS
- KT25 should be a 2 for the Assistant and the HOD – when deployed, you may have the responsibility of witnessing contractor trials
- LT40 MS LCMM is listed as a 2 – it should be changed to a 3 to mimic the NCS IOS
- LT 60 MS MSEO is listed as a 3 and should be changed to a blank
- LT 80 in both MS and NCS for the Assistant, Jr Proj Eng and Jr Eng SO should be a 2
- MT25 and MT30 MS Assistant is rated a 3 and should be a 2.
- Section M LCMM scoring is not compatible between NCS and MS. The scoring should reflect the NCS IOS except MT35, LCMM should be a 3 vice a 2
- MT 45 should be blanked for Assistant and HOD since on board ship, the HOD is not involved in the CP process, only in managing a budget, which is covered under MT50 CS. The QSPWB recommends that MT 45 change from a 2 to a 3 at the Jr Eng SO level. The basics of CP are taught in BOQ ashore (phase VI), achieving a level 2.

Annex C - Integrated Occupation Specifications - Knowledge Value Changes

- K0235 MS – Scoring should be a 2 for all positions to reflect the NCS scores
- Remove K0495 NCS/ K0540 MS
- K0680 NCS/ K0730 MS should increase to a 2 for A/CSEO and up and should become a 3 for the Snr Eng. Reword from "Submarine safe-to-dive" to "SUBSAFE" policy.
- Add a knowledge item for Submarine Materiel Certification. The scoring should be Level 2 until Snr Eng, where it is then increased to a 3
- Note that the above 2 items would impact the BOQ qualification
- K1205 – CSEO knowledge level should be a 3 while the Jr Eng, Jr Eng SO and LCMM should be a 2
- K1225 –Change scoring to a 1 for Asst and CSEO/MSEO
- K1230 – Scoring for the different positions should be re-examined
- K0720 MS – Recommend that the value be increased to a 3 for MSEO
- K0825 MS – Recommend that the value be increased to a 3 for MSEO
- K0810 MS – Recommend that the value be increased to a 2 for Asst, 3 for MSEO
- K0880 MS - Recommend that the value be increased to a 3 for MSEO
- K1055, K1060 MS – Recommend that the value be increased to a 2 for all groups
- K1055, K1060, K1070, K1075 MS/K1115-1135 NCS need their scores examined and where/when this knowledge is increased.
- K2005 MS – Recommend that the value be decreased to a 1 for Asst and MSEO
- K2010-K2085 MS: Recommend that the value be increased to a 3 for Asst
- K2090 MS – This knowledge is a duplicate of K0035, it should be deleted
- K2515, K2520, K2535, K2540 MS – There is no equivalent knowledge in the NCS IOS, it needs to be added
- K2525, K2530, K2545 MS – These should be removed because they are already being covered under specific knowledge areas
- K2555, K2560 MS – These should be removed because it is referencing the knowledge of a form
- K2565-2590 MS – Recommend that the value be increased to a 3 for Asst
- K2595, K2615 MS - There is no equivalent knowledge in the NCS IOS, it needs to be added
- K2620 MS - Recommend that the value be increased to a 3 for Asst
- K2630 MS - Recommend that the value be increased to a 3 for Asst. Also, it should be added to the NCS IOS
- K2635-2650 MS – Recommend deletion since they are OOD qualifications and not part of this phase of training
- K2655-2665 MS - No equivalent knowledge in the NCS IOS, needs to be added
- K2670 MS – The wording is vague, it should be reworded and then added to the NCS IOS
- K2675 MS - There is no equivalent knowledge in the NCS IOS, needs to be added

- K2680 MS – Recommend that the value be decreased to a 2 for MSEO and add the knowledge to the NCS IOS
- K2685-2705 MS - There is no equivalent knowledge in the NCS IOS, needs to be added

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	2.0
Date: 16 June 2023	**Who Initiated Change:** HMCS Venture
Requirement for the Amendment: Rewrite of the QSP into the new format. No changes to content were made	
Description of Changes: New format of the QSP, no changes to content in chapters 1 and 2	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. There are no resources, equipment, supplies, and other elements required to support this training.

ANNEX F – COSTING INFORMATION

This course has no cost as it is an OJPR to complete while posted to a sea-going vessel as the assistant head of department.

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This course will take and estimated 365 days to complete the OJPR and complete the final board.

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
Entire QSP	All POs	To assess trainee skills and knowledge to stand as a naval combat systems engineering head of department	Board-style assessment	Upon completion of OJPR, and 365 days as assistant head of department	Indicate if the time identified is the amount of time per student or the total time to complete the EC	Board room at local TE	Nil	See annex I	Failure will require TRB with extension and reboard if deemed necessary

ANNEX I – ASSESSMENT CHECKLISTS

The MS HOD board is a technical and professional board. Therefore, the board's conduct shall be the pinnacle of professionalism.
Proper uniform, professional conduct, accurate terminology, and personal confidence are all vital to the candidate's success.
Commonly, the first two questions are completed (each of which must be individually passed) at which time the board will take a break to discuss carrying on. If the candidate is performing acceptably, the board will continue. If the candidate has failed either of the first two questions, the board may be ended and a debrief given. Actions upon failure of a board are IAW NAVORD 4500-20.
The AISE board consists of 5 questions, with the following recommended schedule and timings,

Q	Portion	Time (mins)	Morning Board 0800 Start	Afternoon Board 1300 Start
	Candidate presentation preparation	30	0730-0800	1230-1300
	Candidate and Board Member Introduction	10	0800-0810	1300-1310
1	Technical Failure	60	0810-0910	1310-1410
2	Engineering Drill	25	0910-0935	1410-1435
	Break	20	0935-0955	1435-1455
3	Damage Control	25	0955-1020	1455-1520
4	Technical Administration & Policy	25	1020-1045	1520-1545
5	Divisional Administration & Leadership	25	1045-1110	1545-1610
	Board Deliberation / Marking	15	1110-1125	1610-1625
	Initial Feedback and Completion	5	1125-1130	1625-1630

Question 1 | Technical Failure
30 minutes to prepare. 60 minutes to present, including brief and questions.
A brief to the Commanding Officer regarding a failure scenario and the courses of action (CoAs) available, including an analysis of each one and the recommended CoA. Candidate is to brief on the incident, the initial actions taken and the operational impact of the failure. Use of the risk assessment process IAW the relevant NAVORDS and other references is expected. Candidates are to illustrate their follow-on actions and what organizations they will engage to assist or provide support. The brief should include a risk assessment of the CoAs in accordance with NAVORD 3001-1, with the recommended CoA being that for which the reduction in risk (materiel, personnel and environmental) is as low as reasonably practicable (ALARP) in consideration of the resulting mission capability. The candidate shall note all reasonable assumptions made during the CoA development. The brief shall include follow on steps and a plan to implement and support the selected CoA including support and administration from shore facilities and other RCN departments.

Question 2 | Engineering Drill
25 minutes, including response and follow up questions.
Preparing, scheduling, conducting and assessing engineering drill periods are a key part of the MSEO duties. There are two parts to this question. In part one the candidate is given an administrative question about the conduct and assessment of engineering drills as it relates to training and readiness of the MSED. The manual of emergency procedures; marine systems engineering, is the source reference for these questions.
Part two consists of an engineering difficulty or emergency in a scenario-based situation. The candidate is given a scenario and drive mode configuration, and then a trigger alarm or incident is given. Like the AIRY qualification board, the candidate must respond with the correct assessment of the situation, expected initial actions for key personnel and what is required to maintain propulsion and prevent damage. The candidate should provide a brief to the CO as they would in a real event. Additionally, the candidate is expected to provide a plan for initial troubleshooting and discuss possible plant limitations and suspected or possible cause for the engineering problem.

Question 3 | Damage Control
5 minutes to prepare and 20 minutes to present including follow up questions.
A damage control question, requiring the candidate to respond or describe the response to an emergency at sea. The candidate will be given a short time to properly compose an answer and then present to the board. A static state board or projected BDCS shall be provided to the candidate. The candidate should discuss the ship's internal battle, including any relevant policies, ship specific SOPs and considerations. The candidate should discuss the status of personnel and equipment during close-up of action stations, and while fighting a fire or flood, as it relates to the question. The candidate should also discuss the current stability, displacement, and structural integrity considerations and potential impacts.

Question 4 | Technical Administration & Policy
5 minutes to prepare and 20 minutes to present including follow up questions.
A technical administration, policy or operations question. This question can be one of any number of questions focusing on technical policy (e.g. work periods, engineering changes, etc.) or other form of policy such as docking procedures or technical management. The candidate shall have a detailed knowledge of the content and application of relevant policies and orders as well as an ability to discuss administrative resources and requirements from shore establishments. The Fleet

Maintenance Facility, Formation Technical Authority, Fleet Engineering Operations, Fleet Engineering Readiness, Maritime Equipment Program Management, Ship-borne Helicopter Operating Procedures, Ship's Standing Orders, Naval Engineering Manual, Naval Materiel Management System, Technical Bulletins, and Technical Advisories are all within the scope of this question.

Question 5 | Divisional Administration & Leadership
5 minutes preparation, 15 minutes delivery, 5 minutes for follow up questions.
This question does not duplicate Leadership Respect Honour material, but is concerned with training progression for departmental personnel, disciplinary and administrative policy, and personnel management within a department. Candidates are to be given time to compose an answer. This question is used to assess the candidate's readiness to assume the role of Head of Department on a warship beyond technical policy. A strong awareness of corporate culture, ethos, and deck plate leadership is required. Question five focuses on the application of policy and the candidate's ability to not only discuss applicable policy, but also how to interpret and apply the policies in a situationally appropriate manner.

The following tables are used in the evaluation of the candidates during the performance check.

Candidate:	**Ship:**	
Qualification: **MS Eng HOD (AISE)**	**Date:**	
Chairperson:	**Examining Officer:**	**Command Representative:**
HOD:		

Instructions:
Each question is scored out of 10 for content and presentation which are weighted to create a score out of 10 for each question.[1] Each question is then weighted to create an overall board mark.[2] The candidate shallachieve a minimum overall board mark of 60%, including a minimum of 60% for questions 1 and 2 on bothcontent and presentation (critical questions).

Score	**Content**	Score	**Presentation**[3]
0	Unable to answer any part of the question or even begin to discuss the topic.	0	Many major deficiencies
1	Unable to communicate a basic level oftechnical knowledge of engineering principles.	1	Some major deficiencies
2	Able to answer minor parts of the main question but unable to answer follow upquestions.	2	Many intermediate deficiencies
3	Displays a very low level of technical knowledge of engineering principles.	3	Some intermediate deficiencies
4	Able to answer parts of the main questionand a limited number of follow up questions displaying only a restricted level of technical knowledge.	4	Many minor deficiencies
5	Requires assistance answering the main question and requires prodding to answerfollow up questions.	5	Some minor deficiencies
6	Displays an acceptable level of technicalknowledge of engineering principles and administration requirements.	6	Acceptable
7	Displays an above average level of technical knowledge of engineering principles.	7	Good
8	Displays a high level of technical knowledge of engineering principles.	8	Very Good
9	Generates discussion through the displayof excellent technical knowledge of engineering principles.	9	Excellent
10	Displays an exceptional level of technicalknowledge answering the main question and follow up questions well beyond theexpected level.	10	Exceptional

Note 1	Qualification	Content	Presentation
	~~NEI~~	~~0.8~~	~~0.2~~
	~~BOQ~~	~~0.8~~	~~0.2~~
	HOD	0.6	0.4

Note 2	Question	Weight
	1	5
	2	P/F
	3	4
	4	3
	5	3

NTO Qualification Board Marking Sheet

CANDIDATE:			**SHIP:**		
QUALIFICATION: MS Eng HOD (AISE)			**DATE:**		
			CCA REP USE ONLY		
QUESTION	**CONTENT**	**PRESENTATION**	**QUESTION TOTAL**	**WEIGHT FACTOR**	**WEIGHTED SCORE**
1				**5**	
2				**P/F**	
3				**4**	
4				**3**	
5				**3**	

MARK (%): $\frac{\sum Weighted\ Score}{2} =$

RESULT: **PASS / FAIL** (circle one)

COMMENTS:

NOTE: Name and signature are mandatory

Name of Board Member _____ Signature

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)
(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 2.0	31 / 05 / 2023	OJPR Formatted into new RCN QSP format, original OJPR from 2011

Name and Initials	Rank	Service Number

Date Training Started	
Date of Completion	

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

1. Progress review shall be carried out quarterly by the Divisional Officer and the Training Officer.

OJPR Guidance

1. The preferred method of training for this package is to participate in the evolutions. However, if this is not at possible due to the ship's program, a written or oral report is an acceptable alternative. The format for the report can be established by the MSEO as they see appropriate to the situation. For skills, such as planning for different types of missions outside what is programmed, the trainee should be given various scenarios to present as a briefing or in the format of a table top exercise to provide maximize training experience.

2. The activity of "Perform as the MSEO" only appears in PO 021, however, many of the other POs' requirements may be met while acting in the position. It is compulsory that the candidate be given the opportunity to act as the HOD for a period of one month including managing the activities of a short work period. It is also compulsory that they be afforded the opportunity to brief the Commanding Officer ten times while alongside and fifteen times while at sea. The requirements of each PO should be reviewed carefully to ensure that the trainee is provided the opportunity to act in the various roles as outlined.

3. Risk management is a key element in the day-to-day responsibilities of the HOD. As such, it has been included as a PO in itself; however, the principles learned in that PO must be applied throughout the OJPR.

Definitions

1. Within the OJPR, there are several definitions that require further clarification. They are:

a. to include – this means including but not limited to;

b. such as – examples to clarify the intent of the requirement, not a comprehensive list;

c. briefing – a briefing should include the technical and administrative details and where appropriate, the overall impact on the ship's program and on other departments;

d. mission readiness – this includes technical and personnel readiness associated with any tasking that the unit could be assigned; and

e. to the satisfaction of the MSEO – means to a standard that demonstrates a level of knowledge and managerial competence, including command perspective, appropriate to the challenge of being a HOD at sea.

2. A master copy of this OJPR will be kept and maintained by the supervisor. A duplicate copy may be made for retention by the trainee, for the purpose of recording and reviewing individual progress.

3. As the trainee completes each Performance Objective (PO) to the required standard, the Training Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

4. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise available. This will be indicated by marking an “E” (Exception) in the remarks column of the OJPR and including a brief explanation opposite that PO.

5. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under “Additional Duties” at the end of the OJPR.

6. Upon completion of the OJPR sheet (trainee has completed all required POs), the following action will be taken:

a. Qualification Board:

(1) in accordance with NAVORD 4500-20, nominate the trainee for the qualification board;

(2) provide, to the board convening authority representative, a completed copy of the OJPR sign off sheet in accordance with the timings directed by the board convening authority; and

(3) upon successful completion of the qualification board, the convening authority will grant the qualification and will send a message to the career manager and the candidate’s parent unit indicating the date that the qualification was awarded;

b. Administrative (all candidates):

(1) the Commanding Officer shall complete the Certificate of Qualification in this OJPR;

(2) the individual’s Qualification Record Sheet (QRS) shall be completed as described in A-P9-000-015/PT-000. Update action within the MITE module of Peoplesoft is a formation responsibility, currently delegated to CFNES for MARLANT and CFFSE for MARPAC;

(3) the OJPR shall be retained on the individual’s UER until update action is reflected on the Personal Record Resume; and

(4) The NTO Afloat Training Report (Annex G) shall be completed by the candidate's HOD upon completion of the qualification board, serving the same function as a course report would for a course held ashore.

On-Job Performance Record Sheet

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YY YY	OPI
PO 001 Manage Departmental Personnel					
EO 001.01	Advise on Operational Limitations due to Personnel Shortages	Complete para 3a			MSEO
		Complete para 3b			XO
EO 001.02	Manage Individual Training and Education	Written Report for 3a			MSEO
		Complete para 3b through 3e			MSEO
EO 001.03	Provide Divisional Support for military Personnel	Complete para 3			XO
		Written RW and C&P			MSEO
		Briefings on para 3c			MSEO
EO 001.04	Manage operational training	Written Report for para 3a			MSEO
		Complete para 3b			MSEO
PO 002 Liaise with Scientific establishments and Industry					
EO 002.01	Liaise with Scientific establishments and Industry	Written Report for para 3			MSEO
PO 003 Manage Departmental Financial Resources					
EO 003.01	Manage Departmental Financial Resources	Complete para 3a			MSEO
		Written Report for			MSEO
		para 3b through 3g			SYO
PO 004 Manage Shipboard Risk					
EO 004.01	Promote environmental awareness onboard ship	Complete para 3			MSEO
EO 004.02	Manage interdepartmental safety procedures	Complete para 3a			UGSO
		Complete para 3b			XO
EO 004.03	Perform as Unit Environmental Officer	Complete para 3a			MSEO
		Complete para 3b through 3d			MSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
		Written Report for 3e through 3i			MSEO
EO 004.04	Perform the duties of the head of Fuel Spill Response Team	Complete para 3			MSEO
EO 004.05	Ensure compliance with policies/procedures governing handling, stowage and disposal of hazardous materials	Written Report for para 3			MSEO
PO 005 Prepare Technical Documentation					
EO 005.01	Complete routine engineering reports	Complete para 3			MSEO
EO 005.02	Review and maintain technical directives, orders and records	Complete para 3			MSEO
EO 005.03	Prepare maritime technical policy	Complete para 3			MSEO
PO 006 Participate in Interdepartmental Operation Planning					
EO 006.01	Participate in planning ship's operations, program and/or routine	Briefings on para 3			MSEO
EO 006.02	Liaise with CSEO to assess impact of marine systems on combat systems	Written Report for para 3			MSEO
		Written Report for para 3			CSEO
PO 007 Manage Signature Management Program					
EO 007.01	Participate in Ship's Signature Management	Complete para 3			MSEO
PO 008 Manage Shipboard Configuration					
EO 008.01	Control configuration management onboard ship	Complete para 3			MSEO
PO 009 Manage Test and Trials Programme					
EO 009.01	Manage Ship's Test and Trials Programme	Complete para 3			MSEO
		Written Report for para 3			MSEO
		Briefings on para 3			MSEO
PO 010 Manage Readiness					

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
EO 010.01	Maximize marine systems availability within the stated notice for power	Complete para 3			MSEO
		Written Report para 3a ii to 3a iv and 3a ix			MSEO
		Briefings on para 3			MSEO
		Briefings on para 3			MSEO
		Brief the CO 10 Times			MSEO
EO 010.02	Maximize marine systems availability at sea	Complete para 3			MSEO
		Written Report para 3a ii through 3a v			MSEO
		Briefings on para 3			MSEO
		Briefings on para 3			CSEO
		Brief the CO 15 Times			MSEO
EO 010.03	Assess status of DC states	Written Report para 3			MSEO
EO 010.04	Advise on shipboard Degrees of Readiness	Written Report para 3			MSEO
PO 011 Manage Defect Rectification					
EO 011.01	Direct and supervise repair action due to equipment failures	Complete para 3			MSEO
		Written Report for para 3e			SYO
		Briefings on para 3			MSEO
PO 012 Manage Battle Damage					
EO 012.01	Describe the Ship's Damage Control Organization	Written Report for para 3			MSEO
EO 012.02	Direct DCHQ1, DCHQ2 and Section Base Teams	Complete para 3			MSEO
		Act as MSEO during Action Stns 3 Times			MSEO
		Briefings on directing DCHQ and SBTs			MSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
EO 012.03	Manage Damage Control Training	Complete para 3			MSEO
		Create Battle Problem			MSEO
		Create Battle Problem			CSEO
		Create Battle Problem			OPSO
PO 013 Respond to non-routine operations					
EO 013.01	Direct the Fuel Spill Response Team	Complete para 3 for a Fuel Spill Response Exercise			MSEO
EO 013.02	Respond to shipboard engineering emergencies	Written Report for para 3			MSEO
		Briefings on responses to engineering emergencies			MSEO
PO 014 Manage Docking					
EO 014.01	Describe types of docks, including capabilities and limitations of each	Written Report for para 3			MSEO
EO 014.02	Describe the occasions and regulations for docking	Written Report for para 3			MSEO
EO 014.03	Describe the preparation required for docking	Written Report for para 3			MSEO
EO 014.04	Describe weight controls and safety considerations while docked and the regulations/preparations for undocking	Written Report for para 3			MSEO
EO 014.05	Conduct a docking/undocking	Written Report for para 3			MSEO
		Briefings on docking and undocking			MSEO
PO 015 Manage Ship's Tiered readiness Program (TRP)					
EO 015.01	Project manage Departmental	Written Report for para 3			MSEO

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
	responsibilities during a TRP	Briefings on TRP Responsibilities			MSEO
		Briefings on TRP Responsibilities			OPSO
PO 016 Manage Formation Level Maintenance Activities					
EO 016.01	Prepare and staff first line O&M documentation	Complete para 3			MSEO
EO 016.02	Liaise with Formation staff with respect to priority	Complete para 3			MSEO
		Briefs on various scenarios			MSEO
EO 016.03	Monitor contract Work	Briefs on various scenarios			MSEO
EO 016.04	Participate in the TEMMIS programme	Complete para 3			MSEO
PO 017 Manage a Short Work Period					
EO 017.01	Project manage a SWP	Complete para 3			MSEO
		Written report para 3			MSEO
		Act as MSEO during SWP			MSEO
		Briefings on managing a SWP			MSEO
PO 018 Manage ADM(Mat) Level Maintenance Activities					
EO 018.01	Plan a refit/DWP	Complete para 3			MSEO
		Written report para 3			MSEO
		Briefings on managing a refit/DWP			MSEO
EO 018.02	Monitor contract work	Complete written report para 3			MSEO
		Briefs on various scenarios			MSEO
EO 018.03	Provide support to QAR	Complete written report para 3			MSEO
		Briefs on various scenarios			MSEO
PO 019 Manage Ship's Stability					

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
EO 019.01	Manage Ship's Stability	Complete para 3			MSEO
		Briefings on ship's stability			MSEO
PO 020 Manage Liquid Load					
EO 020.01	Ensure safe handling and stowage of fluids	Complete written report para 3			MSEO
EO 020.02	Control Liquid Transfer	Complete para 3			MSEO
		Complete written report para 3a to 3c			MSEO
PO 021 Manage Departmental Systems					
EO 021.01	Describe Main Propulsion and Ancillary Systems	Complete para 3			MSEO
EO 021.02	Describe Auxiliary Systems	Complete para 3			MSEO
EO 021.03	Describe Control Systems	Complete para 3			MSEO
EO 021.04	Describe Electrical Systems	Complete para 3			MSEO

Progress Review Record

Authority & Period	Reviewed By	Signature	Date
Divisional Officer (Q1)			
Training Officer (Q1)			
Divisional Officer (Q2)			
Training Officer (Q2)			
Divisional Officer (Q3)			
Training Officer (Q3)			
Divisional Officer (Q4)			
Training Officer (Q4)			

TRAINING COMPLETION CERTIFICATE

I certify that_____has completed training to the standards stated in QSP – MS Eng - AHOD, except as noted, and is recommended to challenge the Head of Department Qualification Board.

Date training completed:	
MSEO	Date
XO	Date
CO	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Naval Combat Systems Engineering Assistant Head of Department QSP..

TE Cmdt/CO	Date

ANNEX K – TT PLAN

As this is an OJPR, no TT plan was implemented.

ANNEX L – RECORD OF DECISIONS TPWB

1. The original TPWB was sat during the QSWB, which the record can be found in annex D. It has not been duplicated in this annex.

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	**2.0**
Date:12 June 2023	**Who Initiated Change:** HMCS Venture
Requirement for the Amendment: Update to new RCN QSP format, and update to all boards for NCS Eng and MS Eng	
Description of Changes: QSP was formatted into new format, no content changed from the previous QSP in chapters 3 and 4. In annex I, the board was changed to require the candidate to draw during the board.	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

ACRONYM / ABBREVIATION	TERM / DEFINITION
AA	Anti-Air
AAW	Anti-Air Warfare
AWW	Above Water Warfare
ACM	Acoustic Countermeasures
ADM(Mat)	Assistant Deputy Minister (Material)
AER	Automated Establishment Report
ARPS	Acoustic Range Prediction System
ASE	Automotive Service Excellence Standards
ASME	American Society of Mechanical Engineers
ASTN	Switched Transport Network Standards
ASuW	Anti-Surface Warfare
ASW	Anti-Submarine Warfare
ASWEX	Anti-Submarine Warfare Exercise
ATE	Automated Test Equipment
BITE	Built-In Test Equipment
C&P	Counselling and Probation
CbtO	Combat Officer
CCP	Contract Change Proposal
CCS	Command and Control System
CEPA	Canadian Environmental Protection Act
CFAD	Canadian Forces Ammunition Depot
CFAO	Canadian Forces Administrative Orders
CFM	Contractor Furnished Material
CFQAR	Canadian Forces Quality Assurance Representative
CFSS	Canadian Forces Supply System
CFTO	Canadian Forces Technical Order
CHOD	Chief of Department
CIWS	Close-In Weapons Systems
CM	Configuration Management
CM	Corrective Maintenance
CMIS-S	Computerized Maintenance Information System – Ship’s
CRR	Combat Readiness Report
CSAT	Combat System Alignment Test
CSE	Combat Systems Engineering

CSEOTI	Combat System Engineering Officer Technical Instruction
CSEOTM	Combat System Engineering Officer Temporary Memorandum
CTWSC	Combat Team Weapons Safety Certification
CWI	Continuous Wave Illuminator
DAOD	Defence Administrative Orders and Directives
DAPM	Directorate Ammunition Program Management
DC	Damage Control
DCHQ	Damage Control Headquarters
DCOTT	Damage Control Organization Team Training
DDL	Dynamic Deliverable List
DMS	Defence Management System
DoDS	Department of Defence Standards
DRDC	Defence Research and Development Canada
DRE	Defence Research Establishment
DWP	Docking Work Period
EC	Engineering Change
ECP	Engineering Change Proposal
EDWP	Extended Docking Work Period
EHM	Equipment Health Monitoring
ELD	Equipment Logistic Directives
EMC	Electromagnetic Compatibility
EMC	Expenditure Management Course
EMI	Electromagnetic Interference
EMP	Electromagnetic Pulse
EMREQ	Emergency Requirements
EMS	Environmental Management System
EOOW	Engineering Officer of the Watch
EOTI	Engineering Officer Technical Instruction
EPM	Environmental Protection Management
ERT	Emergency Repair Team
ESL	Equipment Support List
Eval	Evaluation
EW	Electronic Warfare
EWP	Extended Work Period
EXDIR	Exercise Directive
FAA	Financial Administration Act
FAA	Formation Ammunition Authority
FDU	Fleet Diving Unit

FEL	Fitted Equipment List
FSR	Field Service Representative
FTA	Formation Technical Authority
FTB	Fleet Technical Bulletin
GSM	Government Supplied Material
HAZMAT	Hazardous Material
HIFR	Helo In Flight Re-fuelling
HOD	Head of Department
HPR	High Priority Requirement
HVAC	Heating, Ventilation, and Air Conditioning
IEEE	Institute of Electrical and Electronics Engineers
IFF	Identify Friend or Foe
ILS	Integrated Logistic Support
IPT	In-Port Training
IR	Infrared
ISO	International Standards Organization
IT&E	Individual Training and Education
LCC	Life Cycle Costing
LCMM	Life Cycle Material Manager
LSA	Logistic Support Analysis
MAAL	MARLANT Ammunition Allowance List
MAF	Maintenance Action Form
MRS	Maintenance Repair Specification
MARCORD	Maritime Command Order
MARE	Maritime Engineer
MARLANTORD	Maritime Forces Atlantic Order
MARPACORD	Maritime Forces Pacific Order
MARPOL	International Convention for the Prevention of Pollution from Ships
MASIS	Material Acquisition and Support Information System
MCAAL	Maritime Command Ammunition Allowance List
MILSTDS	Military Standards
MMIs	Material Management Instructions
MPAAL	MARPAC Ammunition Allowance List
MRC	Mission Readiness Checks
MRSL	Maintenance and Repair Specification List
MSE	Marine System Engineering
NaMMS	Naval Maintenance Management System
NBC	Nuclear, Biological, Chemical
NDE	Non-Destructive Examination
NDE	Non-Destructive Testing

NDID	National Defence Index and Documentation
NEI	Naval Equipment Index
NEM	Naval Engineering Manual
NES	National Exposure Standard
NETE	Naval Engineering Test Establishment
NFP	Notice for Power
NTDS	Naval Tactical Data System
OCCAP	Oil and Coolant Condition Analysis Program
OEM	Original Equipment Manufacture
OHSA	Occupational Health and Safety Act
OPDEF	Operational Deficiency
OPSCHED	Operational Schedule
ORO	Operations Room Officer
OT	Operational Training
OTT	Operational Team Training
PEL	Periodic Engineering Letter
PIF	Pre-Installation Report
PL	Project Leader
PM	Preventative Maintenance
PM	Project Management/Manager
PMRS	Particularised Maintenance Repair Specification
PSL	Particularized Specification List
PWGSC	Public Works and Government Services Canada
QA	Quality Assurance
QAR	Quality Assurance Representative
QC	Quality Control
QETE	Quality Engineering Test Establishment
QHM	Queen’s Harbour Master
QR&O	Queen’s Regulations and Orders
R&D	Research and Development
R&O	Repair and Overhaul
R&SP	Readiness and Sustainment Policy
RADHAZ	Non-Ionizing Radiation Hazard
RAM	Reliability, Availability and Maintainability
RCM	Reliability Centred Maintenance
REMAR	Report on the Management of Resource
RF	Repair Facilities
RSP	Readiness Support Program
RTI	Request for Technical Information
RW	Recorded Warning
SAR	Search and Rescue

SBT	Section Base Team
SCA	Supply Customer Account
SHC	Stock Holding Code
SMMISM	Ship's Maintenance Management Information System Manual
SOCD	Statement of Operational Capability Deficiency
SOP	Standard Operation Procedure
SOR	Statement of Requirements
SOT	System Operability Test
SOW	Statement of Work
SPATE	Special Tools and Test Equipment
SPDET	Safety Program Development and Evaluation Technique
SPF	Sonar Performance Figure
SRB	Senior Review Board
SRI	Sea Readiness Inspection
SS	Ship's Staff
SSE	Ship Systems Engineering
SSO	Ship's Standing Officers
SSRWC	Standard Ship Refit Work Catalogue
SWOAD	Ship Without Air Detachment
SWP	Short Work Period
SYO	Supply Officer
TEMMIS	Test Equipment Maintenance Management Information System
TEWA	Threat Evaluation Weapons Assignment
TMA	Target Motion Analysis
TRANREQ	Transfer Requirement
TSOR	Technical Statement of Requirements
UCR	Unsatisfactory Condition Report
VA	Vibration Analysis
VSTI	Victoria Special Technical Instructions
WASAT	Weapons Alignment System Acceptance Test
WRPS	Work Request Priority Schedule
WSB	Watch and Station Bill